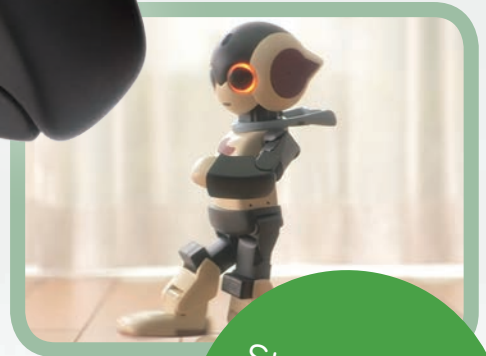
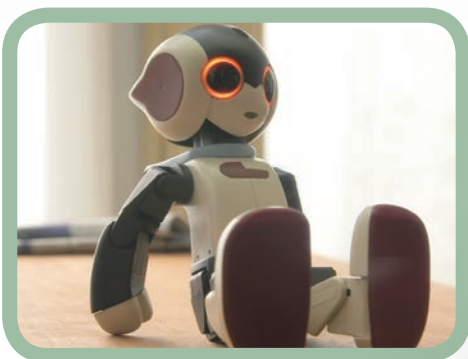


Build your own **Robi**

Pack 11



Stages 39-42:
Receive more of
Robi's electrical
components



Build your own **Robi** CONTENTS

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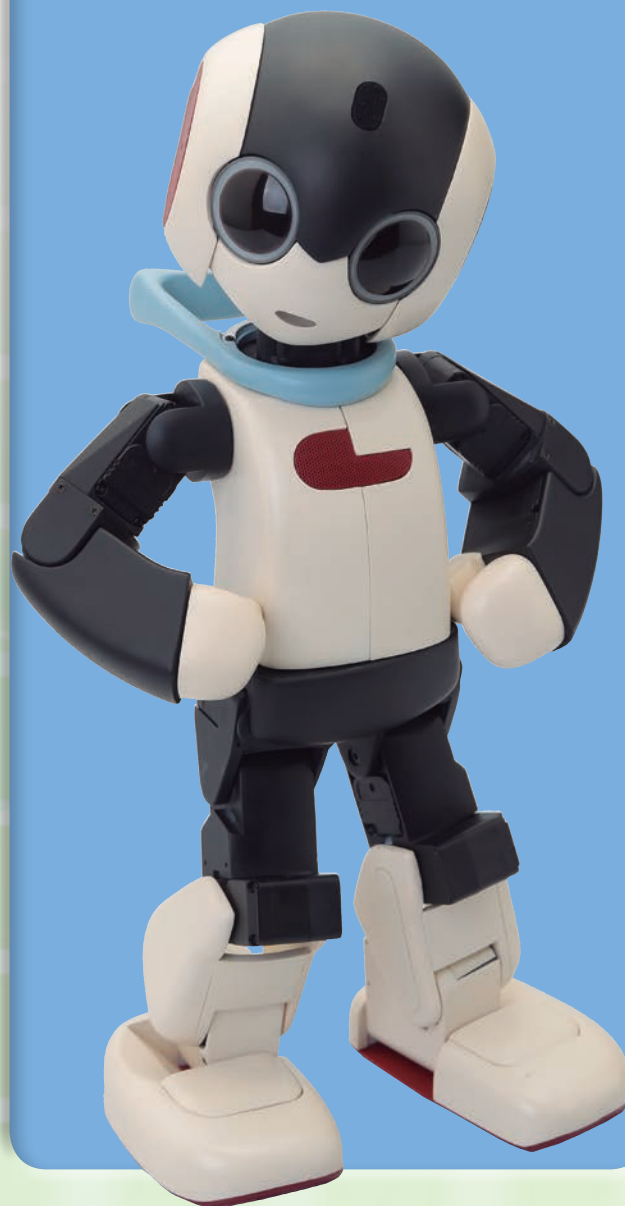
Published in the UK by De Agostini UK Ltd,
Battersea Studios 2, 82 Silverthorne Road,
London SW8 3HE

Published in the USA by De Agostini Publishing
USA, Inc., 121 E. Calhoun Street
Woodstock, IL 60098

Packaged by Continuo Creative,
39-41 North Road, London N7 9DP

Printed in EU

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STAGE 39: ROBI'S CHARGER

The charger supplied with this stage will be used to deliver power to the battery received with the previous pack.

The charger, or adapter, will transfer mains power to Robi's battery, but it is important that it **not** be used or connected to Robi until the power distribution board is installed. It is also

important that you read the safety instructions before using it for the first time, and that the charger is stored safely until then. There are no assembly steps in this stage.



**The charger shown in the step photos throughout the Assembly Guide is a two-pronged European model – but the one you will have received is specific to your region.*



YOUR PARTS

1 Region-specific charger

INSTRUCTIONS

There are no assembly steps in this stage, but remember to read the safety warnings carefully when you receive the charger.

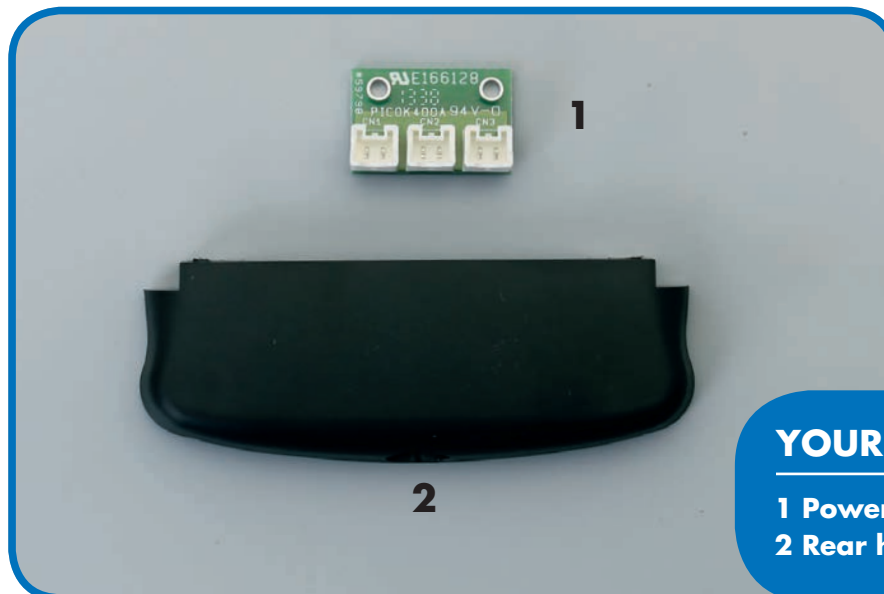


STAGE 40: THE POWER DISTRIBUTION BOARD

In this stage, you have received the power distribution board that will deliver power to Robi's various electronic components on completion, along with the rear hip panel that accommodates the charging socket.

The power distribution board will divide power between the battery, switch and charging cables as required, and is an essential component for the smooth running of Robi on completion – although it will also be used in the testing of

Robi's arms and legs in coming stages. Until then, the power distribution board, along with the rear hip panel also received with this stage, should be stored away safely. Again, there are no assembly steps in this stage.



YOUR PARTS

- 1 Power distribution board
- 2 Rear hip panel

SAFE STORAGE

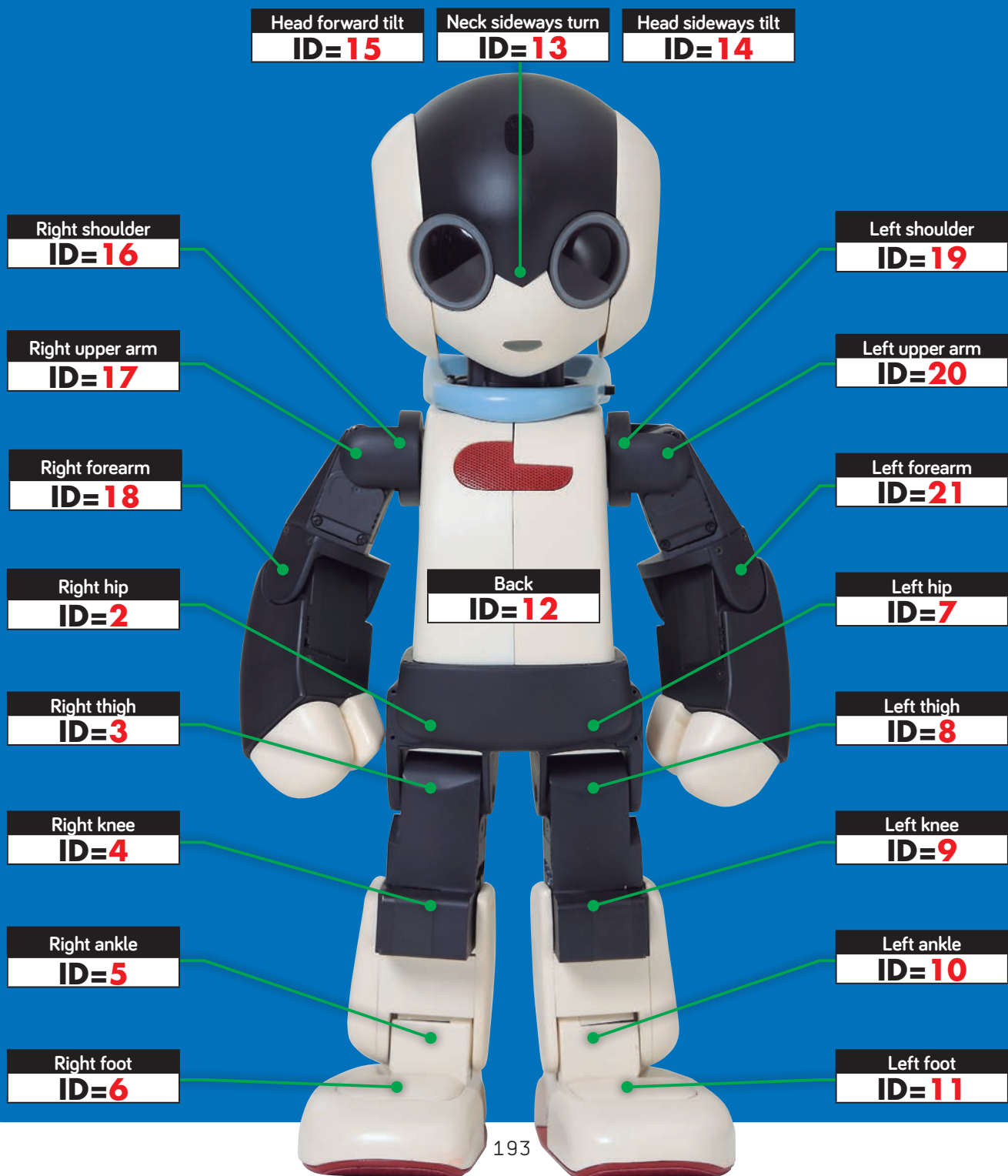
The power distribution board is small and delicate, so has been delivered in bubble-wrap packaging. Keep the part in this wrapping and store it away until it is needed.



SERVO ID NUMBERS



As you will be well aware of by now, each of Robi's servo motors has a unique ID number, which you have been setting using the test board over the previous stages. If one is set with the wrong ID number, Robi will not work properly, and you will have to go back and disassemble the parts holding the servo, disconnect it and then reset it – which can be a fiddly and time-consuming process! The diagram below (also found in Stage 8) will serve as a quick-reference point to help you remember exactly which servo goes where.

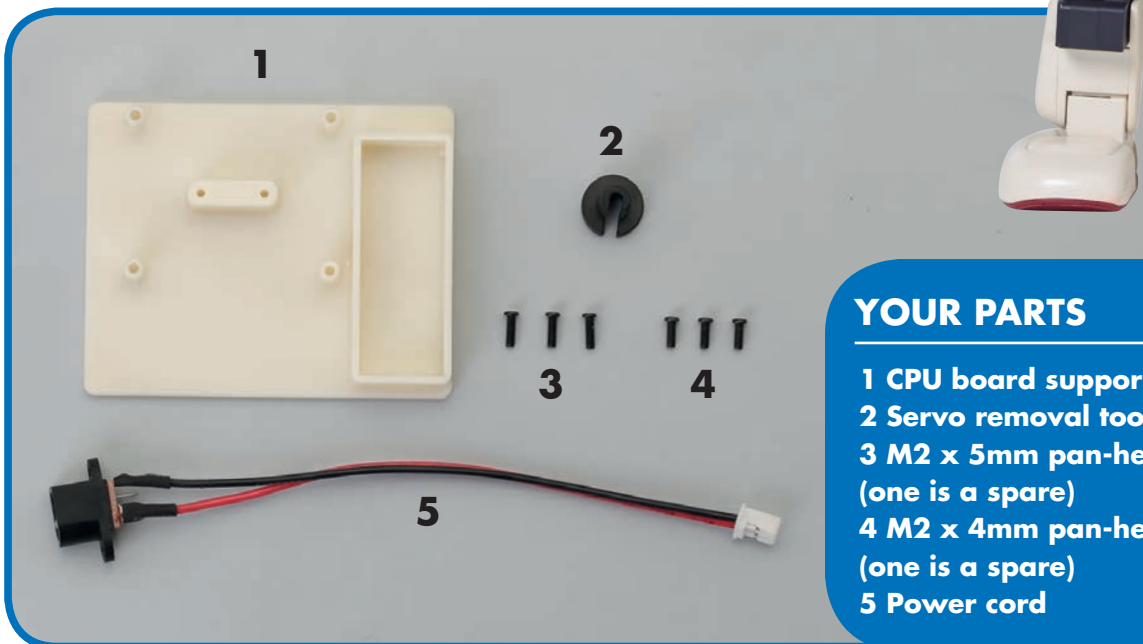


STAGE 41: MOUNT THE BATTERY

In this stage you have received the CPU board support, which will hold Robi's Central Processing Unit on completion, along with the power cord that will carry mains power from the charger to the battery pack.

The CPU board support will hold the Central Processing Unit neatly within Robi's body, and will feed electronic messages to each of the electronic components, such as his limbs, via servo cables to tell these how to

operate. There is a slot on the support to hold the battery pack, which will power Robi. Once you have fitted the power cord to the rear hip panel, you will charge the battery for the first time.



YOUR PARTS

- 1 CPU board support
- 2 Servo removal tool
- 3 M2 x 5mm pan-head screws x 3 (one is a spare)
- 4 M2 x 4mm pan-head screws x 3 (one is a spare)
- 5 Power cord

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Battery
(Stage 38)



Rear hip panel and power
distribution board
(Stage 40)



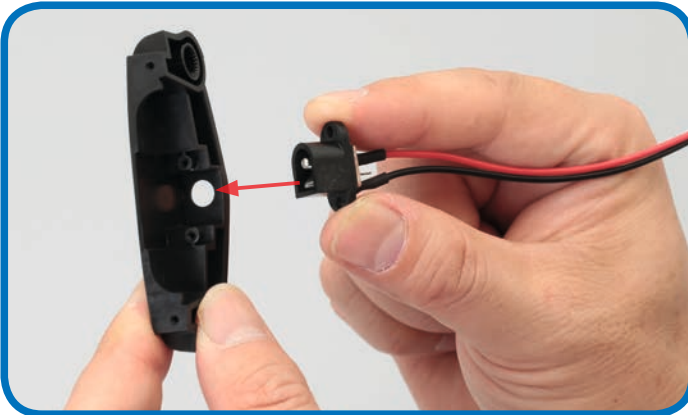
Protective pads
(Stage 3)



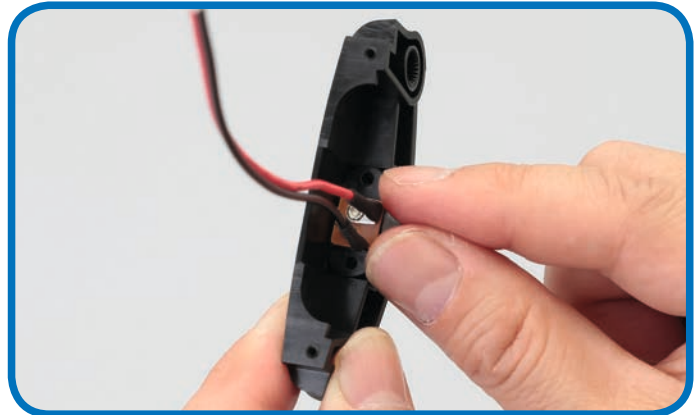
Charger
(Stage 39)



FITTING THE POWER CORD



- 1** Line up the power cord's connector to the hole in the rear hip panel, from the inside.



- 2** Press the connector into the hole so that it sits flush with the inside face of the hip panel.

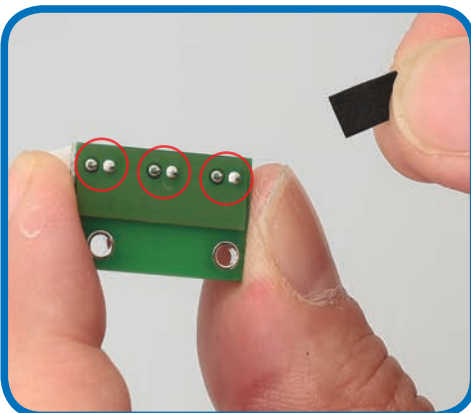


- 3** The jack inside the power cord's connector must be set straight and in the centre of the hole, as shown.

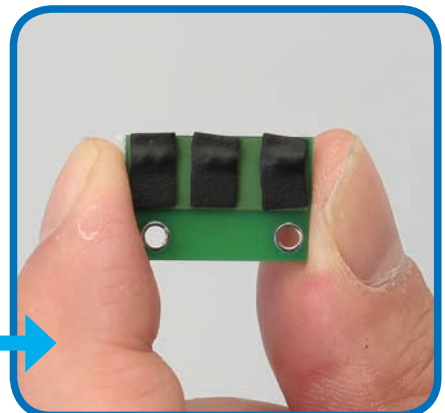
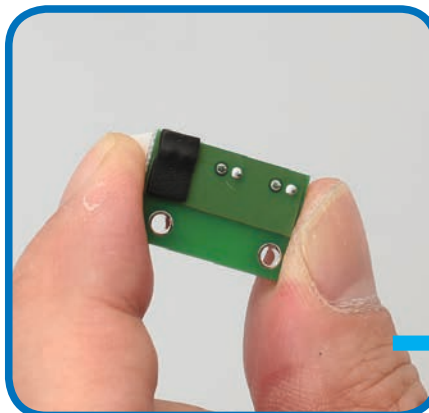


- 4** Secure the connector using two M2 x 5mm pan-head screws.

FITTING PROTECTIVE PADS

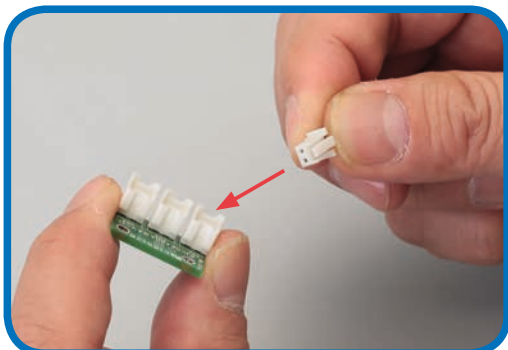


- 5** Hold the power distribution board as shown, then peel off a protective pad from the sheet and line it up to the first of the circled metal solder joints, on the underside of the board.

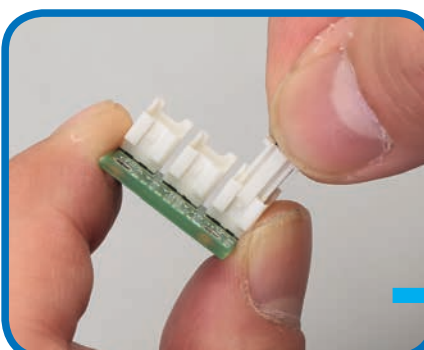


- 6** Repeat for the other two sets of solder joints, so that all are covered. The pads should remain neatly within the board, without any part of them sticking out beyond it.

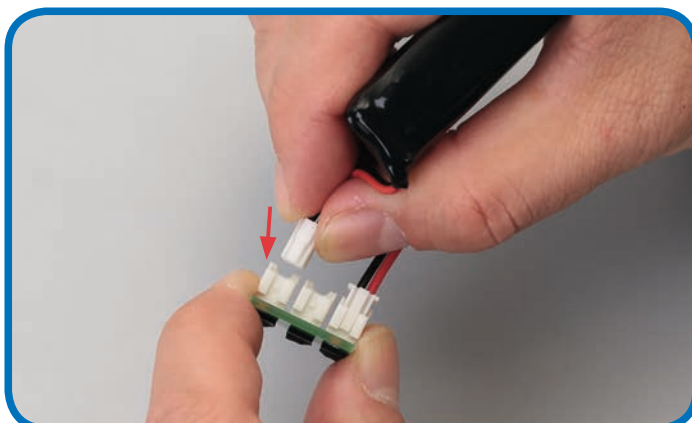
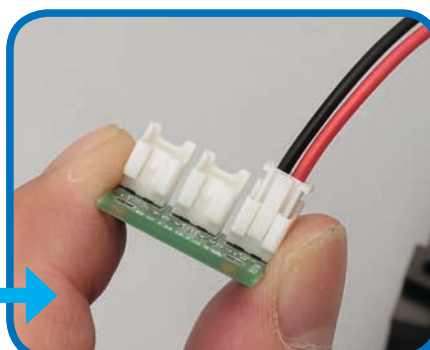
JOINING THE POWER CORD AND BATTERY



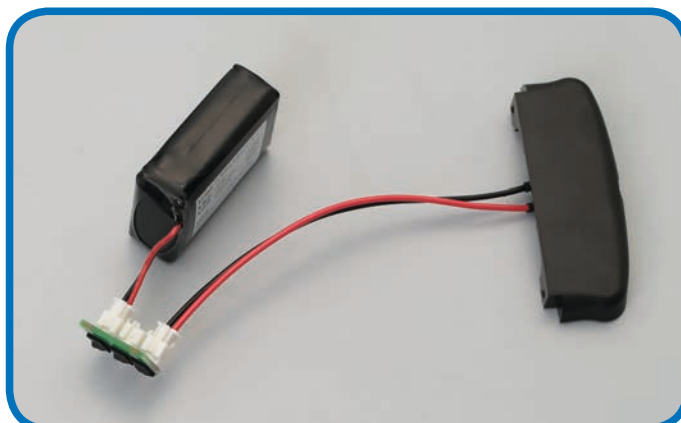
7 Take the white connector at the end of the power cord (leading to the rear hip panel) and line this up to the indicated socket on the power distribution board.



8 Fit the connector's tip into the socket, then, making sure it is perfectly straight, press it into place. The part should be securely held within the socket when this is done, as shown in the right photo.



9 Now line up the white connector at the end of the cables leading from the battery to the socket at the far side of the power distribution board.



10 Press this securely into place as you did in Step 8. The battery and power cord are now connected.

CHARGING THE BATTERY



11 Plug the charger supplied with Stage 39 into the charging port in the rear hip panel, leading to the power cord's connector within.



12 Push this into the socket – it is OK if a few millimetres of metal can still be seen.

ATTENTION!



As with any electrical item, care must be taken when connecting to mains electricity. Never handle the charger with wet hands, and only switch the socket on when the charger is installed. This charger should only be used in a single socket, rather than a multiple or split one, such as those on a 4-way extension lead. Once the charge is complete, remove from the power immediately – this will be after approx. 20 minutes for the first charge, and roughly an hour thereafter.



- 13** Plug the charger into a mains socket. The LED should light up red.

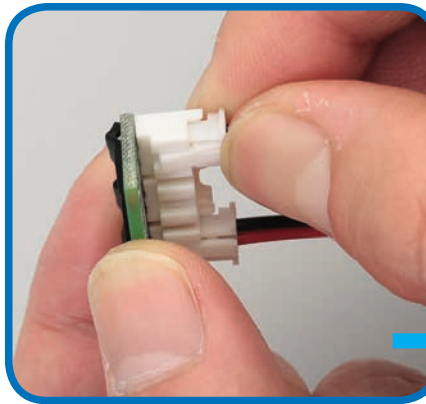


- 14** When the battery is fully charged, the LED will turn green. Turn off the power and remove the charger at this point.

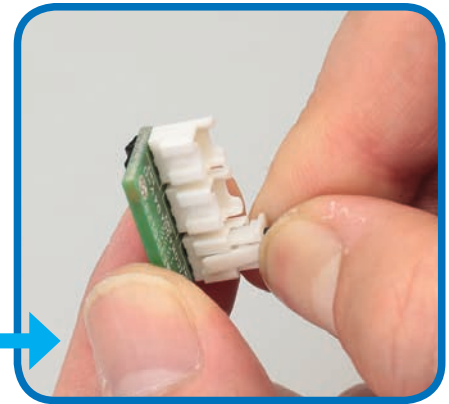
DISCONNECTING THE CHARGER



- 15** Unplug the charger from the charging port in the rear hip panel.



- 16** Pinch the power cord connector to release it from the board, then do the same for the battery's one.



- 17** Put the power distribution board back in its protective bubble wrap bag for safe-keeping.



Completed rear hip panel

The rear hip panel that holds the charging port is now prepared, you have seen how the power distribution board is used and the battery has had its first full charge.

STAGE 42: THE TEST SD CARD

In this stage, you have received the SD card that will be used to test Robi's movement, which will begin in the coming stages.

The micro SD card supplied with this stage is not the one that will control Robi when he is completed, but is specifically programmed to run a series of actions to test the movement of Robi's arms, legs, hip and neck. This is an important process, as it is vital that all the various servos that give Robi his movement are programmed correctly before any further assembly work takes place.

The motion test will take place over a number of assembly stages (starting in the next pack), and will begin with you testing the already-built right arm, left arm and right leg, before completing the process by testing the left leg, hip and neck as these are assembled.

The test SD card is very small and easily lost, so be very sure to keep it safely until it is needed.



YOUR PARTS

1 Test SD card



COMING IN PACK 12

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 43

Receive Robi's CPU board and begin the movement test

PARTS PROVIDED

The CPU that will control Robi's movements on completion.



Stage 44

Prepare Robi's left foot servo and test the left arm

PARTS PROVIDED

Another of Robi's servos.



Stage 45

Build up Robi's left foot and test its servos

PARTS PROVIDED

The left foot cover and sole, along with fixing screws.



Stage 46

Assemble Robi's left lower leg then test his right leg

PARTS PROVIDED

Left lower leg panels, a servo cable and fixing screws.